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HOME PLOTS APPS FIGURE FORMAT TOOLS

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FILE

MATLAB Drive

Files

Workspace

Name	Value	Size	Class
AllocatedSlots	3	1x1	double
FreeSlots	7	1x1	double
TotalSlots	10	1x1	double
Utilization	30	1x1	double

untitled3.m x untitled4.m x untitledexp2.m x untitled5.m x

```
1 clc; clear; close all;
2 TotalSlots = 10;
3 AllocatedSlots = 3;
4 FreeSlots = TotalSlots - AllocatedSlots;
5 Utilization = (AllocatedSlots/TotalSlots)*100;
6 disp('Frame Utilization (%)')
7 disp(Utilization)
8 figure
9 % Graph 1: Simple Line Plot (SAFE - no bar function)
10 subplot(2,1,1)
11 plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth
12 title('GSM Frame Utilization (Allocated vs Free)')
13 xlabel('Slot Type Index (1=Allocated, 2=Free)')
14 ylabel('Number of Slots')
15 grid on
16 % Graph 2: Pie Chart (SAFE)
```

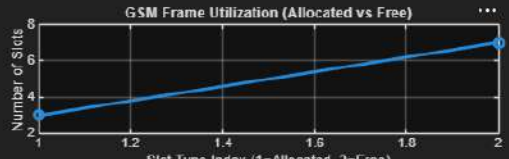
Command Window

Frame Utilization (%)
30

>>

Figure 1 x

GSM Frame Utilization (Allocated vs Free)



Slot Type Index	Number of Slots
1 (Allocated)	3
2 (Free)	7

Slot Distribution in C'II Frame



Category	Count	Percentage
Allocated	3	30%
Free	7	70%

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FILE

MATLAB Drive

Files

Workspace

Name	Value	Size	Class
Alloc	[1,1,1,1,0,0,0,0]	1x8	double
Blocked	[0,0,0,0,0,1,1,1]	1x8	double
TotalSlots	5	1x1	double
Users	[1,2,3,4,5,6,7,8]	1x8	double

```
1 clc; clear; close all;
2 Users = 1:8; % Total users
3 TotalSlots = 5; % Available slots
4 Alloc = zeros(1,8); % Initialize allocation
5 Alloc(1:TotalSlots) = 1; % Assign slots to first use
6 Blocked = 1 - Alloc; % Remaining users blocked
7 disp('User Wise Slot Allocation')
8 disp(table(Users,Alloc,'VariableNames',{'User','Slot Allocation'}))
9 figure
10 % Graph 1: Allocation per user
11 subplot(2,1,1)
12 stem(Users,Alloc,'filled','LineWidth',2)
13 title('TDMA Slot Allocation per User')
14 xlabel('User Index')
15 ylabel('Slot Status (1=Allocated, 0=Blocked)')
16 ylim([-0.2 1.2])
```

Command Window

```
6
7
8
```

Figure 1 x

TDMA Slot Allocation per User

TDMA Slot Utilization Analysis

The image shows the MATLAB software interface. The top menu bar includes HOME, PLOTS, APPS, FIGURE, FORMAT, and TOOLS. The central workspace is divided into three main sections: Files, Command Window, and Figure 1.

Files Section: The left pane shows a list of files in the MATLAB Drive, including untitled3.m, untitled4.m, untitled5.m, untitledexp1.m, and untitledexp2.m. The right pane shows the workspace with variables: Allocate... (5), FreeSlots (5), TotalSlots (10), and Utilization (50).

Command Window: The Command Window displays the output of the script, showing the calculated utilization percentage (50%).

Figure 1: The figure contains two subplots. The top subplot, titled "GSM Frame Utilization (Allocated vs Free)", is a line plot showing the number of slots (Y-axis, 4 to 6) versus the slot type index (X-axis, 1=Allocated, 2=Free). The plot shows a horizontal line at y=5. The bottom subplot, titled "Slot Distribution in C-PI Frame", is a pie chart showing the distribution of slots: 50% Allocated (blue) and 50% Free (yellow).

Script Code:

```
1 clc; clear; close all;
2 TotalSlots = 10;
3 AllocatedSlots = 5;
4 FreeSlots = TotalSlots - AllocatedSlots;
5 Utilization = (AllocatedSlots/TotalSlots)*100;
6 disp('Frame Utilization (%)')
7 disp(Utilization)
8 figure
9 % Graph 1: Simple Line Plot (SAFE - no bar function)
10 subplot(2,1,1)
11 plot([1 2],[AllocatedSlots FreeSlots],'-o','lineWidth
12 title('GSM Frame Utilization (Allocated vs Free)')
13 xlabel('Slot Type Index (1=Allocated, 2=Free)')
14 ylabel('Number of Slots')
15 grid on
16 % Graph 2: Pie Chart (SAFE)
```

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HOME PLOTS APPS FIGURE FORMAT TOOLS

Save As Copy Figure Print Show Code

FILE

MATLAB Drive

Files

Workspace

Name	Value	Size	Class
FrameTime	5.5230	1x1	double
SlotDuration	0.5523	1x1	double
Slots	10	1x1	double
t	1x11 double	1x11	double

```
1 clc; clear; close all;
2 FrameTime = 5.523; Slots = 10;
3 SlotDuration = FrameTime/Slots;
4 t = 0:SlotDuration:FrameTime;
5 figure
6 subplot(2,1,1)
7 stem(t,ones(size(t)),'filled')
8 title('TDMA Time Slot Structure')
9 xlabel('Time (ms)')
10 ylabel('Slot Level')
11 grid on
12 subplot(2,1,2)
13 plot(t,ones(size(t)),'-o','LineWidth',1.5)
14 title('Continuous View of Slot Allocation')
15
16 xlabel('Time (ms)')
17 ylabel('Slot Presence')
```

Figure 1 x

TDMA Time Slot Structure

Continuous View of Slot Allocation

Command Window

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